Anas TBER

Data Science & AI Engineering Student — Centrale Lyon

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Engineering student at Centrale Lyon with hands-on experience in ML, NLP and LLM applications. Seeking an **end-of-studies internship in Data Science or AI** starting **March 2027**.

Education

École Centrale de Lyon — *General Engineering* — *Lyon · Sep. 2023 – Mar. 2027*

- **Courses:** Applied Statistics for Data Science, Machine Learning & Applications, Bayesian Learning & Text Mining, Big Data Technologies, Deep Learning & Applications, Representation & Manipulation of Structured Data, Computer Vision, Probability & Stochastic Processes

Lycée Henri Loritz — *CPGE, PCSI / PSI** — *Nancy · Sep. 2020 – Jun. 2023*

- Intensive preparatory classes in Physics, Mathematics & Engineering Sciences for France's top engineering school entrance exams (Grandes Écoles).

Experience

Data Scientist Intern | BNP Paribas CIB — *Compliance, Anti-Money Laundering (AML)* — *Paris · Jan. – Jul. 2026*

- Delivered a **regulator-approved** network-based AML detection & investigation platform on a **10M+ entity / 40M+ transaction** dataset, from raw ingestion to compliance dashboards; built the pipeline in **Polars / PyArrow** lazy mode with watchlist enrichment
- Designed the detection layer on engineered graph features (centrality, pass-through ratio, flagged-neighbour ratio): **cycle detection** (SCC + DFS) to surface money-mule rings, and an **Isolation Forest** model with **SHAP** explainability ranking suspicious communities with regulator-facing justifications
- Shipped **LLM-powered** features into the **Streamlit** app: one-click AI risk summaries of a community graph and a multi-turn **investigation chatbot** answering free-form questions on an account and its 3-hop neighbourhood (prompt engineering, context-window management)
- **Sole owner and maintainer of the repository** — led the refactor (modular architecture, centralized configuration) and wrote the **full technical documentation** enabling handover to the IT team

Tech & Operations Intern | Gentle Mates — *Professional Esports* — *Paris · Apr. 2025 – Nov. 2025*

- Built an automated social media analytics dashboard (Streamlit/Python) integrating computer vision (**YOLO**) for sponsor logo detection and NLP/**Regex** for brand mention tracking, improving partnership ROI visibility
- Trained a **YOLO** model on a custom-built dataset of sponsor logos to detect partner brands in social media content
- Contributed to the development of impactful software and AI internal tools for analytics, monitoring, and operations

Selected Personal Projects

AirKeys — Camera-Based Gesture Instrument | *Applied ML & Evaluation* — *2026 – Present · Live demo · Code*

- Built a real-time **hand gesture recognition** instrument running fully in the browser: **MediaPipe** landmarks, a self-trained classifier, and on-the-fly personalization via a from-scratch **k-NN**
- Designed a **leakage-safe evaluation pipeline** on 12.5k hands from 9k subjects (**HaGRID**): subject-disjoint splits, deduplication, grouped CV, bootstrap CIs; showed that a single-user model's **99.8%** accuracy fell to **72%** on unseen people, while a model trained on many subjects reaches **98%**
- Currently tackling **open-set rejection** and temporal smoothing, after live-webcam testing showed **99%** gesture accuracy but weak rejection of non-gesture hand poses

arXiv Research Radar — RAG over Recent Papers | *RAG Pipeline* — *2026 · GitHub repo*

- Built an end-to-end **RAG** system answering questions over the latest arXiv papers with per-claim citations: PDF ingestion, **section-aware chunking**, **Sentence-Transformers** embeddings with **FAISS** search, and **Gemini** answers grounded only in retrieved sources (prompt-injection guard)
- Served via **FastAPI + Streamlit** with cost controls (per-IP rate limiting, atomic SQLite daily cap, relevance threshold that skips the LLM on off-topic questions); index refreshed weekly by a **GitHub Actions** pipeline

Skills

Programming	Python, JavaScript, SQL
AI & Data Science	Pandas, Polars NumPy, Scikit-Learn, PyTorch, LightGBM, SHAP, LangChain
Computer Vision	MediaPipe (hand landmarks, gesture recognition), OpenCV
LLM & NLP	RAG (embeddings, FAISS), Gemini/Anthropic/OpenAI APIs, Transformers, Prompt engineering
Web & Viz	FastAPI, Streamlit, Tailwind, Plotly, Pyvis, HTML/CSS
Languages	French (Native), English (Fluent), Arabic (Native), Spanish (Beginner)